

Single Core Copper Conductor, XLPE insulated

Copper Wire Screen and HDPE Sheath
(Cu/XLPE/CW/HDPE).

Voltage Grade

38 / 66 (72.5) KV

Description

Stranded circular or segmental compacted copper conductor Filled with swelling powder between conductor layers, Semi-conductive water blocking tape over conductor to protect the conductor from longitudinal water penetration, semi-conducting layer as conductor screen, XLPE insulated, semi-conducting layer as non metallic insulation screen, Semi-conductive water blocking tape to protect the screen area from longitudinal water penetration, copper wire as metallic insulation screen to withstand the required earth fault current, water blocking tape to protect the screen area from longitudinal water penetration, copolymer aluminum tape to protect the cable from radial water penetration and HDPE sheathed with graphite coating or extruded semi-conducting layer for the D.C Sheath test. Sheath test.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60228, 60840, IEC 60229 and 60811.
Conductor	Plain copper conductor class 2 Circular Comnpacted (applied for C.S.A up to 800 mm2) or Segmental Compacted (Applied from C.S.A = 1000 mm2 and above) Five segments of compacted conductor in sector shape of 72° are assembled together with separation of non-metallic tapes to reduce the skin effect which reduce the Rac.
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded) to the insulation Metallic part: Copper Wires helically applied and banded with a Copper tape to achieve electrical contact Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Outer Sheath	An extruded layer of HDPE shall be applied over the metallic layers for protection against chemical reaction & mechanical protection. Extruded semi conductive layer or graphite coating shall be applied over the outer sheath for D.C testing purpose
Water Penetration layers	i) Longitudinal Water Penetration: 1) the conductor: the conductor layers filled with swelling powder and overlaped semiconductive water blocking tape over the conductor 2) The Metallic part of the insulation screen: An overlaped semiconductive water blocking tape before the metallic screen & an overlaped water blocking tape after the metallic screen ii) Radial Water Penetration: Copolymer aluminum tape applied Longitudinally before the PE outer Sheath
Outer Sheath Color	Black (default color), or any other color according to customer request



Maximum Operating Temp. 90° C

Minimum bending Radius (16-18) x outer diameter

Packing Condition Our standard length on Steel Drum / or according to customer request

Product Code	Nominal Cross Sectional Area	Conductor			Operating Capacitance	Inductance		Thickness of Conductor Screen	Thickness of Insulation	Thickness of Conductor Screen	Thickness of Insulation	Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90 °C			Flat seperated	Trefoil						
													
		mm²	Ω/km	Ω/km		Ω/km	μF/km						
Electrical & Constrctional data (Nominal Values)													
PH-CG20-XB-01-BH	240 R	0.0754	0.0970	0.0973	0.1741	0.6197	0.4348	1.0	13.0	1.0	3.5	61.8	5170
PH-CG21-XB-01-BH	300 R	0.0601	0.0777	0.0781	0.1864	0.6040	0.4191	1.0	13.0	1.0	3.5	64.0	5860
PH-CG22-XB-01-BH	400 R	0.0470	0.0614	0.0619	0.2015	0.5861	0.4012	1.0	13.0	1.0	3.5	66.7	6775
PH-CG23-XB-01-BH	500 R	0.0366	0.0486	0.0494	0.2198	0.5692	0.3843	1.0	13.0	1.0	3.5	70.0	7970
PH-CG24-XB-01-BH	630 R	0.0283	0.0385	0.0396	0.2396	0.5537	0.3689	1.0	13.0	1.0	3.5	73.6	9390
PH-CG25-XB-01-BH	800 R	0.0221	0.0312	0.0326	0.2637	0.5394	0.3546	1.0	13.0	1.0	3.5	78.0	11365
PH-CM26-XB-01-BH	1000 S	0.0176	0.0246	0.0250	0.3100	0.5251	0.3403	1.5	13.0	1.2	3.5	86.9	14270
PH-CM27-XB-01-BH	1200 S	0.0151	0.0218	0.0223	0.3241	0.5187	0.3339	1.5	13.0	1.2	3.5	89.5	15710

Product Code	Nominal Cross Sectional Area	Laying conditions: Trefoil formation			Laying conditions: Flat formation		
		Type of Earthing Bonding System	Direct burial	in Air (Shaded)	Type of Earthing Bonding System	Direct burial	in Air (Shaded)
							
			$p_t=120$ $T = 35\text{ }^{\circ}\text{C}$	$T = 40\text{ }^{\circ}\text{C}$		$p_t=120$ $T = 35\text{ }^{\circ}\text{C}$	$T = 40\text{ }^{\circ}\text{C}$
Continuous current ratings (load factor = 100%) for one circuit in operation (Amperes)							
PH-CG20-XB-01-BH	240 R	Double End Bonded	430	586	Single End Bonded or Cross Bonding	470	678
PH-CG21-XB-01-BH	300 R		481	666		532	778
PH-CG22-XB-01-BH	400 R		541	763		607	902
PH-CG23-XB-01-BH	500 R		606	874		693	1049
PH-CG24-XB-01-BH	630 R		676	996		789	1219
PH-CG25-XB-01-BH	800 R		744	1123		891	1406
PH-CM26-XB-01-BH	1000 S		845	1318		1025	1652
PH-CM27-XB-01-BH	1200 S		892	1378		1099	1790

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or MDPE, LLDPE Sheathed are available according to customer request
The Metallic Screen is designed to carry an earth fault current = 31.5 KA per one second per three phases as default, any other design is also available as per customer request